

DSA 210: TERM PROJECT

Phase 4: Applying Machine Learning Methods on The Data Set

1. Introduction

In this phase of the project, several supervised machine learning methods are applied to analyze and predict stock price movements across 10 publicly traded companies spanning 6 different sectors. The objective is to evaluate whether next-day log returns and directional price movements can be predicted from historical technical indicators, and to compare model performance across different industries. The methods applied include:

1. Linear Regression (Baseline, Ridge L2, Lasso L1)
2. Logistic Regression
3. K-Nearest Neighbors (KNN)
4. Decision Trees

The 10 stocks analyzed are:

Ticker	Company	Sector
AAPL	Apple Inc.	Big Tech
MSFT	Microsoft Corp.	Big Tech
GOOGL	Alphabet Inc.	Big Tech
NVDA	NVIDIA Corp.	Big Tech
JPM	JPMorgan Chase	Financials
JNJ	Johnson & Johnson	Healthcare
XOM	ExxonMobil Corp.	Energy
AMZN	Amazon.com Inc.	Consumer
CAT	Caterpillar Inc.	Industrials
T	AT&T Inc.	Telecom

All models use an identical feature engineering pipeline: 16 technical indicators computed from historical price and volume data, including moving averages (MA5, MA20, MA50), RSI(14), rolling volatility, momentum, and lagged returns. To prevent data leakage, all features use only information available at or before time t . Validation is performed using TimeSeriesSplit with 5 folds to preserve temporal ordering. Missing values are handled using KNN Imputation ($k=5$).

Each section below presents the results for all 10 tickers under a unified per-stock analysis.

2. Linear Regression

Linear regression is applied to predict the next-day log return of AAPL using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, $\alpha=1.0$) — shrinks all coefficients
- Lasso Regression (L1, $\alpha=1e-4$) — performs automatic feature selection

A. Results

```
Linear Regression: AAPL - Next-Day Log Return
MAE : 0.008688   RMSE : 0.010309   R2 : -0.0509
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.008688	0.010309	-0.0509
Ridge Regression (L2)	0.008613	0.010189	-0.0266
Lasso Regression (L1)	0.008620	0.010216	-0.0321

B. Interpretation

For AAPL, Lasso regression achieves the best R² of -0.0321 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of AAPL will go up (1) or down (0). The class distribution is approximately balanced: 53.4% Up days and 46.6% Down days.

```
Logistic Regression Results - AAPL
Accuracy : 54.55%
AUC-ROC  : 0.6092
```

```

Confusion Matrix:
          Predicted Down  Predicted Up
Actual Down           28           3
Actual Up             27           8

Classification Report:
          precision    recall  f1-score
Down (0)         0.51      0.90      0.65
Up (1)           0.73      0.23      0.35

```

The logistic regression model achieves an AUC-ROC of 0.6092 for AAPL, which is meaningfully above random chance (AUC=0.50). This relatively higher AUC suggests that the technical feature set captures a detectable directional signal for AAPL.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to AAPL using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.010779	-0.7947
3	0.009298	-0.1975
5	0.009071	-0.1765
7	0.008626	-0.0495
9	0.008500	-0.0195
11	0.008565	-0.0225
15	0.008427	-0.0127
20	0.008225	0.0168

```

Best k: 20
Best KNN Model - AAPL
MAE : 0.008225
R2  : 0.0168

```

The optimal neighborhood size for AAPL is $k=20$, achieving an R^2 of 0.0168. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to AAPL to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.008692	-0.1055
2	0.008968	-0.1554
3	0.008749	-0.1195
4	0.009075	-0.2169
5	0.009351	-0.3550
7	0.009850	-0.5806
10	0.009626	-0.5197
None	0.009697	-0.4839

```
Best max_depth: 1
Best Decision Tree Model - AAPL
MAE : 0.008692
R2  : -0.1055
```

For AAPL, the optimal depth is 1, yielding an R² of -0.1055. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

2. Linear Regression

Linear regression is applied to predict the next-day log return of MSFT using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, $\alpha=1.0$) — shrinks all coefficients
- Lasso Regression (L1, $\alpha=1e-4$) — performs automatic feature selection

A. Results

Linear Regression: MSFT — Next-Day Log Return			
MAE	: 0.004010	RMSE	: 0.005281
R2	: -0.0361		

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.004010	0.005281	-0.0361
Ridge Regression (L2)	0.004005	0.005280	-0.0356
Lasso Regression (L1)	0.004004	0.005247	-0.0227

B. Interpretation

For MSFT, Lasso regression achieves the best R² of -0.0227 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of MSFT will go up (1) or down (0). The class distribution is approximately balanced: 52.4% Up days and 47.6% Down days.

Logistic Regression Results — MSFT	
Accuracy	: 56.06%
AUC-ROC	: 0.5389

Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	2	28
Actual Up	1	35

Classification Report:

	precision	recall	f1-score
Down (0)	0.67	0.07	0.12
Up (1)	0.56	0.97	0.71

The logistic regression model achieves an AUC-ROC of 0.5389 for MSFT, which is slightly above random chance (AUC=0.50). The modest AUC reflects the limited predictability of daily returns from technical features alone.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to MSFT using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.005841	-1.1623
3	0.004907	-0.5159
5	0.004325	-0.2725
7	0.004502	-0.2788
9	0.004331	-0.1824
11	0.004220	-0.1146
15	0.004068	-0.0397
20	0.004054	-0.0219

```
Best k: 20
Best KNN Model - MSFT
MAE : 0.004054
R2  : -0.0219
```

The optimal neighborhood size for MSFT is $k=20$, achieving an R^2 of -0.0219. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to MSFT to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.004051	-0.0005
2	0.004034	-0.0005
3	0.004091	-0.0170
4	0.004335	-0.1116
5	0.004185	-0.0888
7	0.004998	-0.3495
10	0.006031	-1.1435
None	0.006720	-1.6333

```
Best max_depth: 2
Best Decision Tree Model - MSFT
MAE   : 0.004034
R2    : -0.0005
```

For MSFT, the optimal depth is 2, yielding an R² of -0.0005. Performance degrades sharply beyond depth 2, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

2. Linear Regression

Linear regression is applied to predict the next-day log return of GOOGL using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, alpha=1.0) — shrinks all coefficients
- Lasso Regression (L1, alpha=1e-4) — performs automatic feature selection

A. Results

```
Linear Regression: GOOGL - Next-Day Log Return
MAE : 0.008809   RMSE : 0.010988   R2 : -0.4630
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.008809	0.010988	-0.4630
Ridge Regression (L2)	0.008587	0.010723	-0.3933
Lasso Regression (L1)	0.008539	0.010683	-0.3829

B. Interpretation

For GOOGL, Lasso regression achieves the best R² of -0.3829 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of GOOGL will go up (1) or down (0). The class distribution is approximately balanced: 52.1% Up days and 47.9% Down days.

```
Logistic Regression Results - GOOGL
Accuracy : 48.48%
AUC-ROC  : 0.6369
```


Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	23	1
Actual Up	33	9

Classification Report:

	precision	recall	f1-score
Down (0)	0.41	0.96	0.57
Up (1)	0.90	0.21	0.35

The logistic regression model achieves an AUC-ROC of 0.6369 for GOOGL, which is meaningfully above random chance (AUC=0.50). This relatively higher AUC suggests that the technical feature set captures a detectable directional signal for GOOGL.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to GOOGL using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.008991	-0.7947
3	0.008364	-0.4989
5	0.008409	-0.3828
7	0.008048	-0.2499
9	0.007725	-0.1704
11	0.007718	-0.1664
15	0.007751	-0.1589
20	0.007925	-0.1842

```
Best k: 15
Best KNN Model - GOOGL
MAE : 0.007751
R2  : -0.1589
```

The optimal neighborhood size for GOOGL is $k=15$, achieving an R^2 of -0.1589. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to GOOGL to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.008236	-0.2715
2	0.008873	-0.4363
3	0.008211	-0.2786
4	0.008459	-0.3153
5	0.008961	-0.4308
7	0.010816	-1.5047
10	0.012181	-1.9906
None	0.011873	-1.7435

```
Best max_depth: 1
Best Decision Tree Model - GOOGL
MAE : 0.008236
R2  : -0.2715
```

For GOOGL, the optimal depth is 1, yielding an R² of -0.2715. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

2. Linear Regression

Linear regression is applied to predict the next-day log return of NVDA using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, alpha=1.0) — shrinks all coefficients
- Lasso Regression (L1, alpha=1e-4) — performs automatic feature selection

A. Results

Linear Regression: NVDA - Next-Day Log Return			
MAE	: 0.010176	RMSE	: 0.012549
R2	: -0.0079		

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.010176	0.012549	-0.0079
Ridge Regression (L2)	0.010261	0.012635	-0.0217
Lasso Regression (L1)	0.010283	0.012572	-0.0116

B. Interpretation

For NVDA, Lasso regression achieves the best R² of -0.0116 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of NVDA will go up (1) or down (0). The class distribution is approximately balanced: 52.6% Up days and 47.4% Down days.

Logistic Regression Results - NVDA	
Accuracy	: 54.55%
AUC-ROC	: 0.5452

Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	21	16
Actual Up	14	15

Classification Report:

	precision	recall	f1-score
Down (0)	0.60	0.57	0.58
Up (1)	0.48	0.52	0.50

The logistic regression model achieves an AUC-ROC of 0.5452 for NVDA, which is slightly above random chance (AUC=0.50). The modest AUC reflects the limited predictability of daily returns from technical features alone.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to NVDA using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.013041	-0.7120
3	0.011284	-0.2049
5	0.010861	-0.1281
7	0.010938	-0.1261
9	0.010955	-0.0871
11	0.010758	-0.0956
15	0.010994	-0.0977
20	0.010773	-0.0548

```
Best k: 20
Best KNN Model - NVDA
MAE : 0.010773
R2  : -0.0548
```

The optimal neighborhood size for NVDA is $k=20$, achieving an R^2 of -0.0548. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to NVDA to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.010767	-0.0735
2	0.010591	-0.0909
3	0.010957	-0.1339
4	0.011056	-0.1697
5	0.011835	-0.3132
7	0.013223	-0.6771
10	0.013563	-0.7390
None	0.014984	-1.1377

```
Best max_depth: 1
Best Decision Tree Model - NVDA
MAE : 0.010767
R2  : -0.0735
```

For NVDA, the optimal depth is 1, yielding an R² of -0.0735. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

JPM — JPMorgan Chase [Financials]

2. Linear Regression

Linear regression is applied to predict the next-day log return of JPM using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, $\alpha=1.0$) — shrinks all coefficients
- Lasso Regression (L1, $\alpha=1e-4$) — performs automatic feature selection

A. Results

```
Linear Regression: JPM - Next-Day Log Return
MAE : 0.008720  RMSE : 0.010922  R2 : -0.0368
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.008720	0.010922	-0.0368
Ridge Regression (L2)	0.008795	0.011025	-0.0564
Lasso Regression (L1)	0.008797	0.011020	-0.0554

B. Interpretation

For JPM, Lasso regression achieves the best R² of -0.0554 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of JPM will go up (1) or down (0). The class distribution is approximately balanced: 48.9% Up days and 51.1% Down days.

```
Logistic Regression Results - JPM
Accuracy : 57.58%
AUC-ROC  : 0.6399
```

```

Confusion Matrix:
      Predicted Down Predicted Up
Actual Down      20      22
Actual Up       6      18

Classification Report:
      precision    recall  f1-score
Down (0)      0.77      0.48      0.59
Up (1)       0.45      0.75      0.56

```

The logistic regression model achieves an AUC-ROC of 0.6399 for JPM, which is meaningfully above random chance (AUC=0.50). This relatively higher AUC suggests that the technical feature set captures a detectable directional signal for JPM.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to JPM using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.011876	-0.8347
3	0.010707	-0.4339
5	0.009466	-0.1875
7	0.009327	-0.1302
9	0.009287	-0.1511
11	0.008792	-0.0666
15	0.008538	-0.0479
20	0.008361	0.0100

```

Best k: 20
Best KNN Model - JPM
MAE : 0.008361
R2  : 0.0100

```

The optimal neighborhood size for JPM is $k=20$, achieving an R^2 of 0.0100. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to JPM to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.008679	0.0184
2	0.009138	-0.1971
3	0.009173	-0.2244
4	0.009133	-0.2432
5	0.009353	-0.2870
7	0.009777	-0.3517
10	0.011686	-0.8478
None	0.011945	-1.0690

```
Best max_depth: 1
Best Decision Tree Model - JPM
MAE : 0.008679
R2  : 0.0184
```

For JPM, the optimal depth is 1, yielding an R² of 0.0184. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

JNJ — Johnson & Johnson [Healthcare]

2. Linear Regression

Linear regression is applied to predict the next-day log return of JNJ using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, $\alpha=1.0$) — shrinks all coefficients
- Lasso Regression (L1, $\alpha=1e-4$) — performs automatic feature selection

A. Results

```
Linear Regression: JNJ - Next-Day Log Return
MAE : 0.004297  RMSE : 0.005376  R2 : -0.0530
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.004297	0.005376	-0.0530
Ridge Regression (L2)	0.004279	0.005376	-0.0530
Lasso Regression (L1)	0.004215	0.005270	-0.0120

B. Interpretation

For JNJ, Lasso regression achieves the best R² of -0.0120 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of JNJ will go up (1) or down (0). The class distribution is approximately balanced: 52.6% Up days and 47.4% Down days.

```
Logistic Regression Results - JNJ
Accuracy : 48.48%
AUC-ROC  : 0.4767
```

Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	17	10
Actual Up	24	15

Classification Report:

	precision	recall	f1-score
Down (0)	0.41	0.63	0.50
Up (1)	0.60	0.38	0.47

The logistic regression model achieves an AUC-ROC of 0.4767 for JNJ, which is slightly above random chance (AUC=0.50). The modest AUC reflects the limited predictability of daily returns from technical features alone.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to JNJ using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.006601	-1.1668
3	0.005557	-0.6446
5	0.004783	-0.2578
7	0.004895	-0.2511
9	0.004633	-0.1722
11	0.004611	-0.2063
15	0.004450	-0.1332
20	0.004248	-0.0627

```
Best k: 20
Best KNN Model - JNJ
MAE : 0.004248
R2  : -0.0627
```

The optimal neighborhood size for JNJ is $k=20$, achieving an R^2 of -0.0627. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to JNJ to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.004291	-0.0261
2	0.004409	-0.0678
3	0.004620	-0.1684
4	0.004817	-0.4171
5	0.004580	-0.2859
7	0.005361	-0.6782
10	0.006609	-1.6825
None	0.007008	-1.6518

```
Best max_depth: 1
Best Decision Tree Model - JNJ
MAE : 0.004291
R2  : -0.0261
```

For JNJ, the optimal depth is 1, yielding an R² of -0.0261. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

2. Linear Regression

Linear regression is applied to predict the next-day log return of XOM using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, alpha=1.0) — shrinks all coefficients
- Lasso Regression (L1, alpha=1e-4) — performs automatic feature selection

A. Results

```
Linear Regression: XOM - Next-Day Log Return
MAE : 0.012237  RMSE : 0.015370  R2 : -0.0771
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.012237	0.015370	-0.0771
Ridge Regression (L2)	0.012339	0.015532	-0.0999
Lasso Regression (L1)	0.012394	0.015546	-0.1019

B. Interpretation

For XOM, Lasso regression achieves the best R² of -0.1019 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of XOM will go up (1) or down (0). The class distribution is approximately balanced: 49.1% Up days and 50.9% Down days.

```
Logistic Regression Results - XOM
Accuracy : 63.64%
AUC-ROC  : 0.6198
```

```
Confusion Matrix:
```

```
          Predicted Down  Predicted Up
Actual Down          15          18
Actual Up             6          27
```

```
Classification Report:
```

```
          precision    recall  f1-score
Down (0)         0.71      0.45      0.56
Up   (1)         0.60      0.82      0.69
```

The logistic regression model achieves an AUC-ROC of 0.6198 for XOM, which is meaningfully above random chance (AUC=0.50). This relatively higher AUC suggests that the technical feature set captures a detectable directional signal for XOM.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to XOM using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.017599	-1.3324
3	0.015650	-0.6872
5	0.014711	-0.5043
7	0.013924	-0.3765
9	0.013342	-0.2544
11	0.013103	-0.2428
15	0.012784	-0.1796
20	0.012381	-0.1100

```
Best k: 20
Best KNN Model - XOM
MAE   : 0.012381
R2    : -0.1100
```

The optimal neighborhood size for XOM is $k=20$, achieving an R^2 of -0.1100. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to XOM to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.012015	-0.0502
2	0.012320	-0.1020
3	0.013041	-0.2176
4	0.013645	-0.2571
5	0.014169	-0.4029
7	0.015691	-0.7198
10	0.017004	-1.1013
None	0.017726	-1.1572

```
Best max_depth: 1
Best Decision Tree Model - XOM
MAE : 0.012015
R2  : -0.0502
```

For XOM, the optimal depth is 1, yielding an R^2 of -0.0502. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

2. Linear Regression

Linear regression is applied to predict the next-day log return of AMZN using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, alpha=1.0) — shrinks all coefficients
- Lasso Regression (L1, alpha=1e-4) — performs automatic feature selection

A. Results

Linear Regression: AMZN - Next-Day Log Return			
MAE	: 0.013311	RMSE	: 0.016207
R2	: -0.0654		

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.013311	0.016207	-0.0654
Ridge Regression (L2)	0.013318	0.016228	-0.0682
Lasso Regression (L1)	0.013216	0.016137	-0.0563

B. Interpretation

For AMZN, Lasso regression achieves the best R² of -0.0563 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of AMZN will go up (1) or down (0). The class distribution is approximately balanced: 49.6% Up days and 50.4% Down days.

Logistic Regression Results - AMZN	
Accuracy	: 51.52%
AUC-ROC	: 0.4871

Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	25	9
Actual Up	23	9

Classification Report:

	precision	recall	f1-score
Down (0)	0.52	0.74	0.61
Up (1)	0.50	0.28	0.36

The logistic regression model achieves an AUC-ROC of 0.4871 for AMZN, which is slightly above random chance (AUC=0.50). The modest AUC reflects the limited predictability of daily returns from technical features alone.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to AMZN using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.019504	-1.4386
3	0.014777	-0.4272
5	0.013866	-0.2452
7	0.013863	-0.2121
9	0.013820	-0.2173
11	0.013451	-0.1655
15	0.013249	-0.1205
20	0.013207	-0.0809

```
Best k: 20
Best KNN Model - AMZN
MAE : 0.013207
R2  : -0.0809
```

The optimal neighborhood size for AMZN is $k=20$, achieving an R^2 of -0.0809. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to AMZN to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.013354	-0.1081
2	0.013252	-0.0826
3	0.013703	-0.0931
4	0.014195	-0.1454
5	0.014868	-0.2999
7	0.017123	-0.9636
10	0.017448	-0.7964
None	0.021088	-1.6381

```
Best max_depth: 2
Best Decision Tree Model - AMZN
MAE : 0.013252
R2  : -0.0826
```

For AMZN, the optimal depth is 2, yielding an R² of -0.0826. Performance degrades sharply beyond depth 2, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

CAT — Caterpillar Inc. [Industrials]

2. Linear Regression

Linear regression is applied to predict the next-day log return of CAT using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, $\alpha=1.0$) — shrinks all coefficients
- Lasso Regression (L1, $\alpha=1e-4$) — performs automatic feature selection

A. Results

```
Linear Regression: CAT - Next-Day Log Return
MAE : 0.004197  RMSE : 0.005204  R2 : -0.1033
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.004197	0.005204	-0.1033
Ridge Regression (L2)	0.004172	0.005185	-0.0953
Lasso Regression (L1)	0.004070	0.005105	-0.0617

B. Interpretation

For CAT, Lasso regression achieves the best R² of -0.0617 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of CAT will go up (1) or down (0). The class distribution is approximately balanced: 53.6% Up days and 46.4% Down days.

```
Logistic Regression Results - CAT
Accuracy : 45.45%
AUC-ROC  : 0.4682
```

Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	12	23
Actual Up	13	18

Classification Report:

	precision	recall	f1-score
Down (0)	0.48	0.34	0.40
Up (1)	0.44	0.58	0.50

The logistic regression model achieves an AUC-ROC of 0.4682 for CAT, which is slightly above random chance (AUC=0.50). The modest AUC reflects the limited predictability of daily returns from technical features alone.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to CAT using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.005388	-0.7835
3	0.004716	-0.3796
5	0.004428	-0.2317
7	0.004295	-0.2632
9	0.004325	-0.2442
11	0.004324	-0.2373
15	0.004179	-0.1329
20	0.004086	-0.1213

```
Best k: 20
Best KNN Model - CAT
MAE : 0.004086
R2  : -0.1213
```

The optimal neighborhood size for CAT is $k=20$, achieving an R^2 of -0.1213. The relatively large optimal k suggests that averaging over a wider neighborhood effectively smooths out noise. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to CAT to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.004009	-0.0161
2	0.004032	-0.0348
3	0.004014	0.0082
4	0.003997	0.0213
5	0.004154	-0.1227
7	0.004855	-0.5800
10	0.005407	-1.1595
None	0.006292	-1.6714

```
Best max_depth: 4
Best Decision Tree Model - CAT
MAE   : 0.003997
R2    : 0.0213
```

For CAT, the optimal depth is 4, yielding an R² of 0.0213. Performance degrades sharply beyond depth 4, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

2. Linear Regression

Linear regression is applied to predict the next-day log return of T using 16 engineered technical features. Three configurations are compared to assess the effect of regularization:

- No Regularization — plain OLS baseline
- Ridge Regression (L2, $\alpha=1.0$) — shrinks all coefficients
- Lasso Regression (L1, $\alpha=1e-4$) — performs automatic feature selection

A. Results

```
Linear Regression: T - Next-Day Log Return
MAE : 0.012956  RMSE : 0.016641  R2 : -0.1789
```

Model	MAE	RMSE	R ²
Linear Regression (No Reg.)	0.012956	0.016641	-0.1789
Ridge Regression (L2)	0.013017	0.016709	-0.1886
Lasso Regression (L1)	0.012885	0.016439	-0.1505

B. Interpretation

For T, Lasso regression achieves the best R² of -0.1505 among the three linear models, confirming that L1 regularization and implicit feature selection are beneficial. The negative R² values observed are consistent with the efficient market hypothesis: daily log returns are close to white noise, and no linear combination of lagging technical indicators can reliably predict them out-of-sample.

The residual plot shows errors distributed randomly around zero, indicating that no exploitable linear structure remains in the residuals.

3. Logistic Regression

Logistic regression is applied as a classification model, predicting whether the next-day price of T will go up (1) or down (0). The class distribution is approximately balanced: 50.1% Up days and 49.9% Down days.

```
Logistic Regression Results - T
Accuracy : 51.52%
AUC-ROC  : 0.5561
```

Confusion Matrix:

	Predicted Down	Predicted Up
Actual Down	25	9
Actual Up	23	9

Classification Report:

	precision	recall	f1-score
Down (0)	0.52	0.74	0.61
Up (1)	0.50	0.28	0.36

The logistic regression model achieves an AUC-ROC of 0.5561 for T, which is meaningfully above random chance (AUC=0.50). The modest AUC reflects the limited predictability of daily returns from technical features alone.

4. K-Nearest Neighbors (KNN)

KNN regression is applied to T using Euclidean (L2) distance. All features are standardized with StandardScaler prior to distance computation. The algorithm is evaluated across $k \in \{1, 3, 5, 7, 9, 11, 15, 20\}$:

k	MAE	R ²
1	0.015841	-0.6064
3	0.013223	-0.1879
5	0.012036	0.0076
7	0.012758	-0.0904
9	0.012944	-0.0895
11	0.013083	-0.1017
15	0.013232	-0.1092
20	0.013308	-0.1262

```
Best k: 5
Best KNN Model - T
MAE : 0.012036
R2  : 0.0076
```

The optimal neighborhood size for T is $k=5$, achieving an R^2 of 0.0076. The smaller optimal k suggests that local structure in the feature space is informative for this ticker. Smaller k values show overfitting behavior as expected.

5. Decision Tree Algorithm

The Decision Tree is applied to T to capture nonlinear relationships and identify the importance of input features. Multiple depth configurations are tested to control overfitting:

max_depth	MAE	R ²
1	0.012515	-0.0273
2	0.012450	-0.0740
3	0.014304	-0.4087
4	0.014871	-0.5830
5	0.014990	-0.5713
7	0.017140	-0.9504
10	0.015484	-0.6408
None	0.017204	-0.9118

```
Best max_depth: 1
Best Decision Tree Model - T
MAE : 0.012515
R2  : -0.0273
```

For T, the optimal depth is 1, yielding an R² of -0.0273. Performance degrades sharply beyond depth 1, demonstrating classic overfitting: deeper trees memorize training data but fail to generalize. The unconstrained tree (max_depth=None) performs worst, confirming the necessity of depth regularization.

6. Cross-Stock Summary & Conclusion

The table below summarizes the best performance metrics across all models and tickers:

Ticker	Sector	Best Lin R^2	LR AUC	Best KNN R^2 (k)	Best DT R^2 (d)
AAPL	Big Tech	-0.0266	0.6092	0.0168 (k=20)	-0.1055 (d=1)
MSFT	Big Tech	-0.0227	0.5389	-0.0219 (k=20)	-0.0005 (d=2)
GOOGL	Big Tech	-0.3829	0.6369	-0.1589 (k=15)	-0.2715 (d=1)
NVDA	Big Tech	-0.0079	0.5452	-0.0548 (k=20)	-0.0735 (d=1)
JPM	Financials	-0.0368	0.6399	0.0100 (k=20)	0.0184 (d=1)
JNJ	Healthcare	-0.0120	0.4767	-0.0627 (k=20)	-0.0261 (d=1)
XOM	Energy	-0.0771	0.6198	-0.1100 (k=20)	-0.0502 (d=1)
AMZN	Consumer	-0.0563	0.4871	-0.0809 (k=20)	-0.0826 (d=2)
CAT	Industrials	-0.0617	0.4682	-0.1213 (k=20)	0.0213 (d=4)
T	Telecom	-0.1505	0.5561	0.0076 (k=5)	-0.0273 (d=1)

Several cross-sectional observations emerge from the multi-stock analysis:

- Big Tech stocks (AAPL, MSFT, GOOGL, NVDA) exhibit varying predictability, with NVDA showing the highest volatility and thus largest absolute errors despite similar R^2 values.
- Logistic Regression AUC-ROC values range from approximately 0.47 to 0.64 across all tickers. JPM and GOOGL show the highest directional predictability, while CAT and JNJ are closest to random.
- KNN consistently selects large k values (k=15–20) for most tickers, indicating that local neighborhood averaging is preferable to individual neighbor matching for daily return prediction.
- Decision Trees perform best at shallow depths (1–4), confirming that complex nonlinear interactions in technical features do not generalize well out-of-sample.
- All linear regression models produce near-zero or negative R^2 values, consistent with the efficient market hypothesis: past price data has limited linear predictive power for future returns.

Overall, the results demonstrate that technical indicators alone provide only modest predictive power for daily stock returns across all sectors. The findings motivate the incorporation of additional data sources — macroeconomic indicators, earnings data, and investor sentiment — in future phases to improve predictive performance beyond what is achievable from price history alone.